# Y. Swathi

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## Professional summary

Analytical and detail-oriented Computer Science Engineering student with strong foundations in Machine Learning, Data Science, Statistical Analysis, and Data Engineering workflows. Experienced in handling large datasets, data preprocessing, exploratory data analysis, model building, and visualization using modern analytical tools. Skilled in SQL, Python, and data-driven decision-making, with hands-on exposure to AI/ML applications, predictive modelling. Adept at problem solving, documentation, and communicating insights clearly across diverse audiences.

## Skills

- ***Programming:*** *Python, C, JavaScript, HTML, CSS, SQL*
- ***ML/AI****: NLP, Scikit-learn, TensorFlow, PyTorch*
- ***Data & Analytics****: Data Cleaning, EDA, Statistical Analysis, Predictive Modeling*
- ***Tools and Technologies*** *: MySQL, Power BI, MS Excel, Jupyter Notebook, GitHub,SUMO, TraCI, OpenCV*
- ***Soft Skills:*** *Time Management, Attention to Detail, Confidentiality, Communication, Problem Solving.*

## Education

**SRM Institute of Science and Technology** — Expected June 2026
*Bachelor of Technology in Computer Science Engineering (CGPA: 8.52 / 10.00) till 6th sem* — *Chennai, TamilNadu*

**Vidya Mandir at Estancia** — March 2022
*Non- Medical | 12th | (71%)* — *Chennai,TamilNadu*

**Vidya Mandir at Estancia** — March 2020
*10th | (67%)* — *Chennai,TamilNadu*

## Experience

***Machine Learning Intern – VCODEZ***
*May 2025 - July 2025 | Online*

- *Worked with datasets, performing data cleaning, feature selection, preprocessing, and exploratory data analysis.*
- *Gained exposure to supervised and unsupervised machine learning concepts using scikit-learn.*
- *Practiced building simple ML models for classification and prediction problems.*

***Web Development Intern – Tanmay Sachin Foundations***
*Jan 2024 - Aug 2024 | Online*

- *Learned fundamentals of web development using HTML, CSS, React.js, and MySQL.*
- *Practiced writing basic SQL queries to retrieve and manage structured data.*
- *Assisted in understanding how frontend and backend components interact in web applications.*
- *Followed documentation and task updates under mentor guidance.*

***AI Intern – Teachnook (AICAN Pvt. Ltd.)***
*Jan 2023 - Feb 2023 | Online*

- *Gained hands-on experience in Artificial Intelligence and its real-world applications.*
- *Explored real-world use cases of AI through guided projects and examples.*
- *Learned the basics of AI-driven automation and model workflows.*

## Projects

**PSO-CNN Traffic Optimization System**
*Python, SUMO, TraCI, CNN*

- Worked with simulation-generated large traffic datasets for modelling and prediction.
- Performed feature extraction, data handling, and analysis to evaluate adaptive vs. fixed timing traffic systems.
- Implemented CNN-based vehicle prediction integrated with optimization algorithms.

**Customer Segmentation using K-Means Clustering**
*Python, Pandas, Matplotlib, scikit-learn*

- Performed full analytics lifecycle: dataset creation, preprocessing, cleaning, clustering, and visualization.
- Identified behavioral segments to support data-driven marketing strategies.

**Senior Companion AI Assistant**
*Python, NLP, Streamlit, Twilio*

- Leveraged NLP workflows, speech recognition, and automation for call/SMS reminders.
- Designed dashboards, improved user-centric interaction flow.

## Certifications and Publications

- AWS Academy: AI/ML Foundations
- Fortinet Certified Associate in Cybersecurity
- Generative AI in Action – IBM
- Cisco – Networking Basics
- Research Paper Presented at ICIoT 2025

## Hackathons and Extra curricular

Volunteer work at Seva Chakkara Children's Home (4 weeks) as part of Community Connect program.

Member of Society of Women Engineers (SWE) and IEI Student Chapter at SRM.

Participated in Graph-Q workshop conducted by Alexa Developers and Github SRM community

Participated in E-GNITE'24 Workshop at SRM Institute of Science and technology

Participated in Atrixx Hackathon – ECE Dept, SRMIST

Participated in Ideathon – IEEE SSIT SRMIST SBC

HackSolstice'23 – Manipal Institute of Technology (Virtual Mode)